# SIMATS Online Programming Lab

DS PROGRAMMING

SIMATS ENGINEERING

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QUESTIONS 10 / 10 completed

Single Linked List

## SINGLE LINKED LIST

"A college maintains a list of student roll numbers in the order they are registered. New students may join at different stages of the registration process. Develop a C program using a singly linked list to store the roll numbers and perform insertion operations at the beginning, end, and specified position. Display the updated list after each insertion

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node
{
  int data;
  struct node*next;
};
int main()
{
  struct node*head=NULL,*temp,*newnode;
  int n,i;

  printf("enter number of nodes:");
  scanf("%d",&n);

  for(i=1;i<n;i++);
  {
    newnode=(struct node*)malloc(sizeof(struct node));

    printf("enter data:");
```

OUTPUT

```
enter number of nodes:enter data:linked list:10
```

INPUT

```
3
10
20
30
```

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Deletion Operatior

## DELETION OPERATIONS

"A company maintains employee records in a linked list. Employees may leave the organization, requiring their records to be removed. Write a C program to implement deletion operations from the beginning, end, and a specified position

Test Case 1

Input

List: 10 → 20 → 30

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
#include<stdlib.h>

struct node
{
  int data;
  struct node*next;

};
int main(){
  struct node *head ,*second,*third;

  head=(struct node*)malloc(sizeof(struct node));
  second=(struct node*)malloc(sizeof(struct node));
  third=(struct node*)malloc(sizeof(struct node));

  head->data=10;
  second->data=20;
  third->data=30;

  head->next=second;
```

OUTPUT

```
2030
```

INPUT

```
3
20
30
40
```

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QUESTIONS 10 / 10 completed

Linked List

## LINKED LIST

"A library stores book identification numbers in a linked list. Develop a C program that searches for a given element and displays its position if found.

Test Case 1

Input

List: 10 → 20 → 30 → 40
Search Element: 30

Output

Element found at position 3"

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node
{
int data;
struct node*next;

};
int main(){
  struct node*head,*second,*third;
  struct node*prev=NULL,*curr,*next;

  head=(struct node*)malloc(sizeof(struct node));
  second=(struct node*)malloc(sizeof(struct node));
  third=(struct node*)malloc(sizeof(struct node));

  head->data=10;
  second->data=20;
  third->data=30;
```

OUTPUT

INPUT

40
50
70
80

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QUESTIONS 10 / 10 completed

Singly Linked List-l

## SINGLY LINKED LIST-NODES

A hospital maintains patient records using a linked list. To generate reports, the system needs to determine the total number of records currently stored.Write a C program to count and display the total number of nodes in a singly linked list.

Test Case 1

Input

10 → 20 → 30 → 40 → 50

Output

Total number of nodes

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdio.h>

struct node{
  int data;
  struct node*next;
};
int main()
{
  struct node*head,*second,*third,*temp;
  int count=0;

  head=(struct node*)malloc(sizeof(struct node));
  second=(struct node*)malloc(sizeof(struct node));
  third=(struct node*)malloc(sizeof(struct node));

  head->next=10;
  second->next=20;
  third->next=NULL;
   temp=head;
  while(temp!=NULL)
```

OUTPUT

INPUT

40
59
09
87
65

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QUESTIONS 10 / 10 completed

Single Sorted Link

## SINGLE SORTED LINKED LIST

Two departments maintain separate sorted lists of student IDs. For centralized processing, both lists need to be merged while preserving the sorted order.Write a C program to merge two sorted singly linked lists into a single sorted linked list.

Test Case 1

Input

List 1: 10 → 30 → 50
List 2: 20 → 40 → 60

Output

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node{
  int data;
  struct node*next;

};
int main(){
  struct node*head=NULL,*new,*p;
  int n,x;

  printf("Enter number of nodes:");
  scanf("%d",&n);

  for (int i=0;i<n;i++){
    new=malloc(sizeof(struct node));
    scanf("%d",&new->data);
    new->next=NULL;

    if(head==NULL||new->data<head->data){
```

OUTPUT

```
Enter number of nodes:sorted list:10 20 30 40 50
```

INPUT

```
5
30 10 50 20 40
```

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QUESTIONS 10 / 10 completed

Sorted Linked List-

## SORTED LINKED LIST-2

"A database contains sorted customer IDs, but due to data-entry errors, duplicate records have been created. Develop a C program that removes duplicate nodes from a sorted linked list and displays the updated list.
Test Case 1

Input

10 → 10 → 20 → 20 → 30 → 30

Output

10 → 20 → 30 → NULL"

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node{
  int data;
  struct node*next;
};
int main(){
  struct node *a,*b,*c,*p;

  a=malloc(sizeof(struct node));
  b=malloc(sizeof(struct node));
  c=malloc(sizeof(struct node));

  a->data=10;
  b->data=20;
  c->data=30;

  a->next=b;
  b->next=c;
  c->next=NULL;
```

OUTPUT

```
soarted list:10 20 30
```

INPUT

```
5
30 10 50 20 40
```

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QUESTIONS 10 / 10 completed

Circular Linked List

## CIRCULAR LINKED LIST

"In a round-robin CPU scheduling system, processes are executed cyclically. Such applications can be represented using a circular linked list. Write a C program to create a circular linked list and display all nodes by traversing the list exactly once. Write a C program to create a circular linked list and traverse all nodes exactly once.

Test Case 1

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node{
  int data;
  struct node*next;
};

int main(){
  struct node*head,*a,*b,*p;

  a=malloc(sizeof(struct node));
  b=malloc(sizeof(struct node));

  a->data=10;
  b->data=20;

  head=a;
  a->next=b;
  b->next=head;

```

OUTPUT

```
circular list:10 20
```

INPUT

```
5
10 20 30 40 50
```

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QUESTIONS 10 / 10 completed

Doubly Linked List

## DOUBLY LINKED LIST

"A music player maintains a playlist where users can move both forward and backward between songs. This can be efficiently implemented using a doubly linked list. Write a C program to create a doubly linked list and display nodes in forward and backward directions.

Test Case 1

Input

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
#include<stdlib.h>

struct node{
  int data;
  struct node*prev,*next;
};
int main(){
  struct node*a,*b,*p;

  a=malloc(sizeof(struct node));
  b=malloc(sizeof(struct node));

  a->data=10;
  b->data=20;

  a->prev=NULL;
  a->next=b;
  b->prev=a;
  b->next=NULL;
```

OUTPUT

```
Doubly list:10 20
```

INPUT

```
6
10 20 30 40 50 60
```

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QUESTIONS 10 / 10 completed

Polynomials

## POLYNOMIALS

"In scientific computing applications, mathematical polynomials are often represented using linked lists where each node stores a coefficient and exponent. Write a C program to represent two polynomials using linked lists and perform polynomial addition.

Test Case 1

Input

Polynomial 1: $5x^2$ +

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<stdlib.h>

struct node{
  int coef,pow;
  struct node*next;

};
int main(){
  struct node *a,*b,*c;

  a=malloc(sizeof(struct node));
  b=malloc(sizeof(struct node));
  c=malloc(sizeof(struct node));

  a->coef=5;a->pow=2;
  b->coef=3;b->pow=1;
  c->coef=2;c->pow=0;

  a->next=b;
  b->next=c;
```

OUTPUT

```
polynomial:5x^2+3+1
```

INPUT

```
5
30 60 89 45 76
```